Aashutosh AV

EDUCATION

GEORGIA TECH
Master of Science
Major: Computer Science
Minor: Machine Learning
GPA: 4.0/4.0
Expected May 2027
Atlanta, GA

BITS PILANI
Bachelor of Engineering
Major: Computer Science
Minor: Data Science
GPA: 9.1/10
Graduated May 2025
Pilani, India

SKILLS

PROGRAMMING
Languages:
Python • Java • C++ • SQL • Bash
AI/ML Frameworks:
HuggingFace • PyTorch • OpenCV • W&B • LightGBM

TOOLS & SYSTEMS
Claude Code • Cursor • AWS (EC2, DynamoDB) • Azure • Distributed Systems • Git • HPC

SELECT COURSEWORK
Knowledge-Based AI • Deep Learning • Algorithms • Computer Vision • Graph Learning • ML Theory

LINKS

Email: avaashutosh04@gmail.com
Ph. No: 972.918.1130
LinkedIn: linkedin.com/in/aashutosh-av
Portfolio: aashutoshav.github.io
GitHub: github.com/aashutoshav

HONORS / ACHIEVEMENTS

2025
Institutional Scholarship for Research @ Georgia Tech

2025
Graduated First Class with Distinction @ BITS Pilani

EXPERIENCE

C21U, GEORGIA TECH | Graduate Research Assistant
Jan. 2026 - Present | Atlanta, GA

- Developing autonomous LLM teaching agents for Socratic Minds, GT's production AI tutoring platform.
- Built few-shot GPT-4o pipeline for RAG-assisted ICAP engagement classification; scalable automated dialog coding.
- Built AWS EC2 + DynamoDB eval framework logging real-time traces; quantified Inter-Rater Reliability for human-in-the-loop validation.

HAYS LAB, GEORGIA TECH | Graduate Researcher
Jan. 2026 - Present | Atlanta, GA

- Building NVS-Bench: sparse-view Novel View Synthesis eval framework benchmarking diffusion & feedforward models on ScanNet++.
- Ran context-selection ablations; temporal attention interpretability analysis to diagnose geometric failure modes.

MICROSOFT RESEARCH | Research Intern
Jan. 2025 - July 2025 | Bangalore, India

- Co-invented ORBIT, online bin-packing algorithm for shard placement; 35% footprint reduction across 100K+ Azure VMs, $100M+ projected savings.
- Co-built LUNA: LightGBM quantile-regression forecasting pipeline for peak resource loads under asymmetric loss.
- Co-formulated DRV, a stochastic QoS metric (KDE + bootstrapping) modeling tail-error rates.
- Open-sourced LOADSTAR, a policy-simulation framework benchmarking placement algorithms; cut compute variance ($\ln(P80/P20) \rightarrow 1.34$).

PUBLICATIONS & PATENTS

MEASURING ACTIVE COGNITION THROUGH UPTAKE IN AI-MEDIATED SOCRATIC DIALOGUE | AERA, Under Review • 2027
ICAP-based, RAG-assisted coding shows interactive AI uptake tracks exchange-level reasoning better than behavioral volume.

SCLARO: GROUNDED SCENARIO-LEVEL SCENE UNDERSTANDING | WACV (Core A), Under Review • 2027
Relation-aware VLM with disentangled encoders & EMA distillation; curated Action-Genome Dataset (1.8M pairs).

ANALYZING & OPTIMIZING NOSQL WORKLOADS FOR COSMOS DB | PVLDB Vol. 19 (Core A*) • 2026
Datacenter-scale workload characterization; tail-error prediction & replica placement across 100K+ Azure nodes.

NARRATING FOR YOU: AUDIO-VISUAL NARRATING FACE GENERATION | WACV (Core A) • 2026
Multi-entangled latent space for synchronized joint audio-video talking-face generation from a single image.

WORKLOAD DISTRIBUTION BASED ON PROJECTED ERROR COUNTS | U.S. Patent & Trademark Office • 2025
Error-projection scheme preemptively redistributing NoSQL workloads to prevent QoS violations.

LATENT FLOW DIFFUSION FOR DEEPFAKE VIDEO GENERATION | CVPR (Workshop) • 2024
LFD2G: optical-flow prediction + dual ViT encoding + flow-conditioned diffusion, disentangling identity from motion.